Dummit & Foote: Exercises

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0.1 Basics

For Exercises 1 to 4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathscr{B} = \{X \in \mathscr{A} : MX = XM\}$, where $\mathscr{A}$ denotes the set of 2×2 matrices with real entries.

(1) Determine which of the following elements of $\mathscr{A}$ lie in $\mathscr{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Note that the 1st matrix is M itself so that it belongs to $\mathscr{B}$, since $MX = XM = M^2$. Further note that the 3rd, 5th, and 6th matrices are the zero, identity, and exchange matrices respectively. Then the 3rd and 5th belong to $\mathscr{B}$ while the 6th does not. Then only the remaining matrices to check are the 2nd and 4th matrices. For the 2nd matrix:

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot1+1\cdot1 & 1\cdot1+1\cdot1 \\ 0\cdot1+1\cdot1 & 0\cdot1+1\cdot1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \\ XM &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot1+1\cdot0 & 1\cdot1+1\cdot1 \\ 1\cdot1+1\cdot0 & 1\cdot1+1\cdot1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \end{aligned}$$

Then $MX \neq XM$ and the 2nd matrix does not belong to $\mathscr{B}$. For the 4th matrix:

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1\cdot1+1\cdot1 & 1\cdot1+1\cdot0 \\ 0\cdot1+1\cdot1 & 0\cdot1+1\cdot0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \\ XM &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot1+1\cdot0 & 1\cdot1+1\cdot1 \\ 1\cdot1+0\cdot0 & 1\cdot1+0\cdot1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

So that $MX \neq XM$. □

(2) Prove that if $P, Q \in \mathscr{B}$, then $P + Q \in \mathscr{B}$ where $+$ denotes the usual sum of two matrices.

Solution. For Exercises 2 and 3, let

$$P = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad Q = \begin{pmatrix} e & f \\ g & h \end{pmatrix}, \quad P + Q = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}, \quad P \cdot Q = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

where $PM = MP$ and $QM = MQ$. Then

$$\begin{aligned} M(P+Q) &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e+c+g & b+f+d+h \\ c+g & d+h \end{pmatrix} \\ &= \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix} + \begin{pmatrix} e+g & f+h \\ g & h \end{pmatrix} \\ &= MP + MQ \\ &= PM + QM \\ &= \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix} + \begin{pmatrix} e & e+f \\ g & g+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & a+e+b+f \\ c+g & c+g+d+h \end{pmatrix} \\ &= \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= (P+Q)M \end{aligned}$$

□

(3) Prove that if $P, Q \in \mathscr{B}$, then $P \cdot Q \in \mathscr{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. Using PQ, proceed as we did above with $M(PQ)$. Rewriting the entries will result in $(MP)Q$. Since $P \in \mathscr{B}$, then we have $(PM)Q$. We rewrite entries again to result in $P(MQ)$. Because $Q \in \mathscr{B}$, we have $P(QM)$, and a final rewrite results in $(PQ)M$. □

(4) Find conditions on p, q, r, s which determine precisely when $\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathscr{B}$.

Solution. Let X be the matrix described above. Note that

$$\begin{aligned} MX &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} p+r & q+s \\ r & s \end{pmatrix} \\ XM &= \begin{pmatrix} p & q \\ r & s \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & p+q \\ r & r+s \end{pmatrix} \end{aligned}$$

Because $X \in \mathscr{B}$, then we may compare entries to obtain the following:

$$\begin{cases} p + r = p \\ q + s = p + q \\ r = r \\ s = r + s \end{cases}$$

The first and fourth equations force $r = 0$, and the second equation forces $p = s$. Then $\mathscr{B}$ is classified as

$$\mathscr{B} = \left\{ \begin{pmatrix} p & p+q \\ 0 & p \end{pmatrix} \middle| \; p, q \in \mathbb{R} \right\}$$ □

(5) Determine whether the following functions f are well defined:
  (a) $f : \mathbb{Q} \to \mathbb{Z}$ defined by $f(a/b) = a$.
  (b) $f : \mathbb{Q} \to \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.
  (a) No, because $1/2 = 2/4$, but $f(1/2) = 1$ and $f(2/4) = 2$.
  (b) Yes; suppose $a/b = c/d$. Then $ad = bc$, or that $a^2d^2 = b^2c^2$. Then $a^2/b^2 = c^2/d^2$, or $f(a/b) = f(c/d)$. □

(6) Determine whether the function $f : R^+ \to \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well defined.

Solution. No. Note that $1 = 1.000\ldots = 0.999\ldots$, but $f(1.000\ldots) = 1$ and $f(0.999\ldots) = 9$. □

(7) Let $f : A \to B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f.

Solution. The relation above is an equivalence relation, since $=$ is already an equivalence relation on B.

Consider now some $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. The equivalence class of a is the set $\{x \in A \mid x \sim a\}$. By definition of $\sim$, this is equal to the set $\{x \in A \mid f(x) = f(a) = b\}$, which is precisely the fiber of f over b. □